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Timescapes Next Generation Archive

Implementing a Fedora based system – architecture, design & components

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The problem

The archive platform treats all files as "digital objects" and does not allow the modelling of complex structures of information and its inter-relationships.

It is not possible to clearly display the connections between artefacts produced from a number of interviews and cohort activities over a number of phases.

The solution

The Fedora Commons platform will allow the archive to represent concepts and relationships between concepts, such as collections, waves, and longitudinal case studies to be directly represented in the archive.

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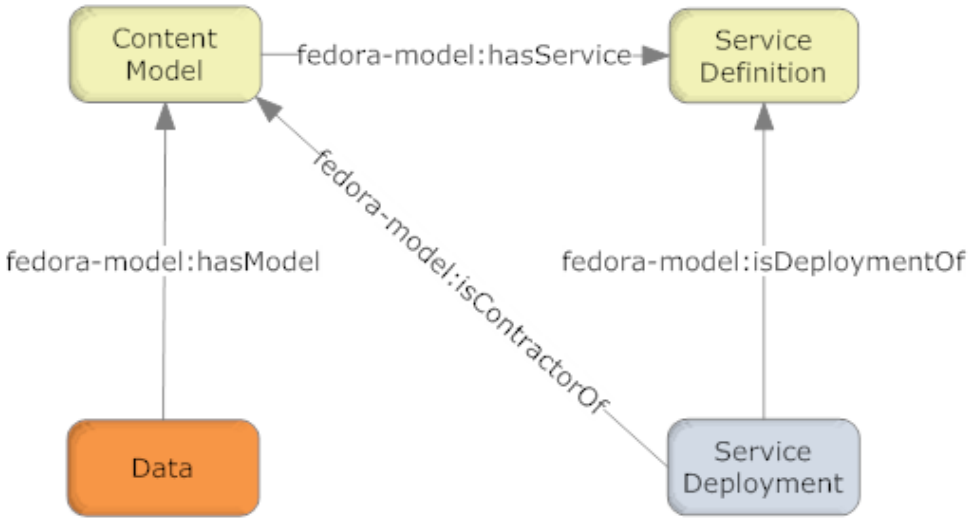
Fedora Content Model Architecture



Content models define what a data object can contain in terms of “data streams”

Data streams are defined as one or more MIME typed objects that can be optional or mandatory

Data objects declare their structure by specifying what content model (s) they have using a relationship – “hasModel”

Dissemination of data streams can be defined using “Services”

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Creating content models

Content Model

ID	Format	Optional
Metadata	XML	False
Interview		
	RTF	False
	PDF	True
	HTML	True

The content model shown describes an object with a metadata data stream and an interview data stream

The metadata data stream is XML, the interview data stream can be RTF, PDF or HTML but there must be an RTF data stream

The data does not have to provide optional data streams, these could be provided by “Services” i.e. convert the RTF stream to PDF when requested

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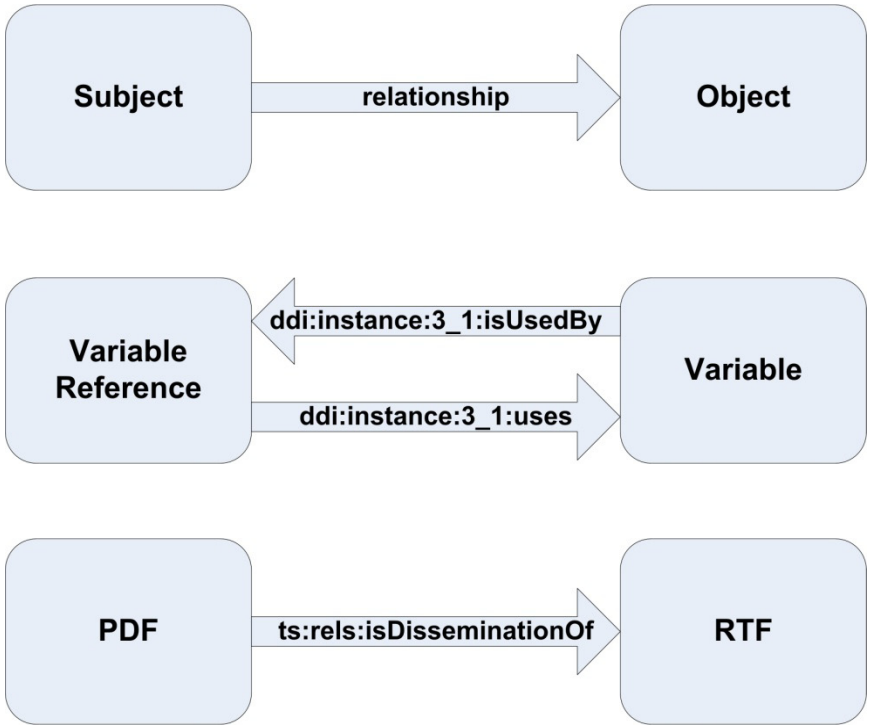
Linking data objects



Relationships are formed between subject and object, they stored in the Mulgara triple store

This store can be queried using SQL like languages

Relationships can be between data objects, data streams within objects and to “literal” values e.g.

<Albert> <hasGender> “Male”

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Using relationships

- Archive
  - Men as Fathers
    - Men as Fathers: Wave Two
      - Brown
      - Green
      - Moore
      - Clarke
      - Evans
      - Morris
      - Lynch

Relationships can be used to model structures such as projects, waves and cases

Here the case “Brown” is related to “Wave Two” using a defined relationship <tsmd:isPartOfWave>

Any ontology can be created to define relationships that are required to model structure, aggregation, reference etc

Timescapes | Brown

Case :	Brown
Description :	Waves One and Two Interviews 2000

Title	Data	Metadata
Wave One Interview 2000		
Wave Two Interview 2000		
Wave Four Interview 2008		
Wave Four Interview context 2008		

Relationships can be used to link data objects to aggregate objects e.g. data files to cases and metadata to data files

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SOLR – searching and browsing

Simple Search

Enter one or more search terms in the box below

brown interview 2008 | Search

Search Results

Title	Data Type	Data File
Men as Fathers: Brown Wave Four Interview Context	InterviewContext	
Men as Fathers: Brown Wave Four Interview 2008	Interview	
Men as Fathers: Brown Waves One, Two and Four Int	Interview	

Page 1 of 1 View 1 - 3 of 3

Advanced Search

Enter one or more search terms in the box below and select the fields you wish to search over

ADD

Project | Case ID | Fieldwork Date | Fieldworker ID | Age | Gender | Data Type | Keyword | Description

Enter search term: brown

And | Or

Project | Case ID | Fieldwork Date | Fieldworker ID | Age | Gender | Data Type | Keyword | Description

Enter search term: interview

Remove

Search

Search Results

Title	Data Type	Data File
Men as Fathers: Brown Waves One, Two and Four Interviews 2000 - 2008	Interview	
Men as Fathers: Brown Wave One Interview 2000	Interview	
Men as Fathers: Brown Wave Two Interview 2000	Interview	
Men as Fathers: Brown Wave Four Interview 2008	Interview	
Men as Fathers: Brown Wave Four Interview Context 2008	InterviewContext	

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Simple searches can be done by just entering one or more search terms

The search will look for data objects that have any of the search terms in pre-configured metadata fields.

Advanced searches can select the metadata fields to be search and the logical operator used to combine the search terms. Multiple entries and allowed to refine or broaden a search and can be removed to amend the search performed so far.

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Faceted Browsing and Searching

593 Items

SUBJECT	LABEL
Male	ts:08afafb7-00ba-48d6-93b8-76047908ea8d1
Fatherhood	ts:08afafb7-00ba-48d6-93b8-76047908ea8d3
Masculinity	ts:08afafb7-00ba-48d6-93b8-76047908ea8d5
Pregnancy	ts:08afafb7-00ba-48d6-93b8-76047908ea8d7
Transitions	ts:08afafb7-00ba-48d6-93b8-76047908ea8d9

Subjects

Context Fatherhood Interview Male Masculinity Pregnancy Transitions context

60 Items filtered from 593 originally (Reset All Filters)

SUBJECT	LABEL
Pregnancy	ts:08afafb7-00ba-48d6-93b8-76047908ea8d7
Pregnancy	ts:192f59d3-4709-4f51-9139-94e3ea07ff297
Pregnancy	ts:196113af-a340-4afd-a00c-41c73269da9b7
Pregnancy	ts:197983f6-98f1-48f3-96fa-c9b0a10531257

Subjects

Context Fatherhood Interview Male Masculinity Pregnancy Transitions context

Default search for all items returns 593 results

Selecting “pregnancy” as a subject filters the search results leaving 60 items

Further subjects can be selected/de-selected to refine expand the view of the search results

The filters applied using faceted browsing can be added to the default search to create a new search whose results could be further refined using the subject filters

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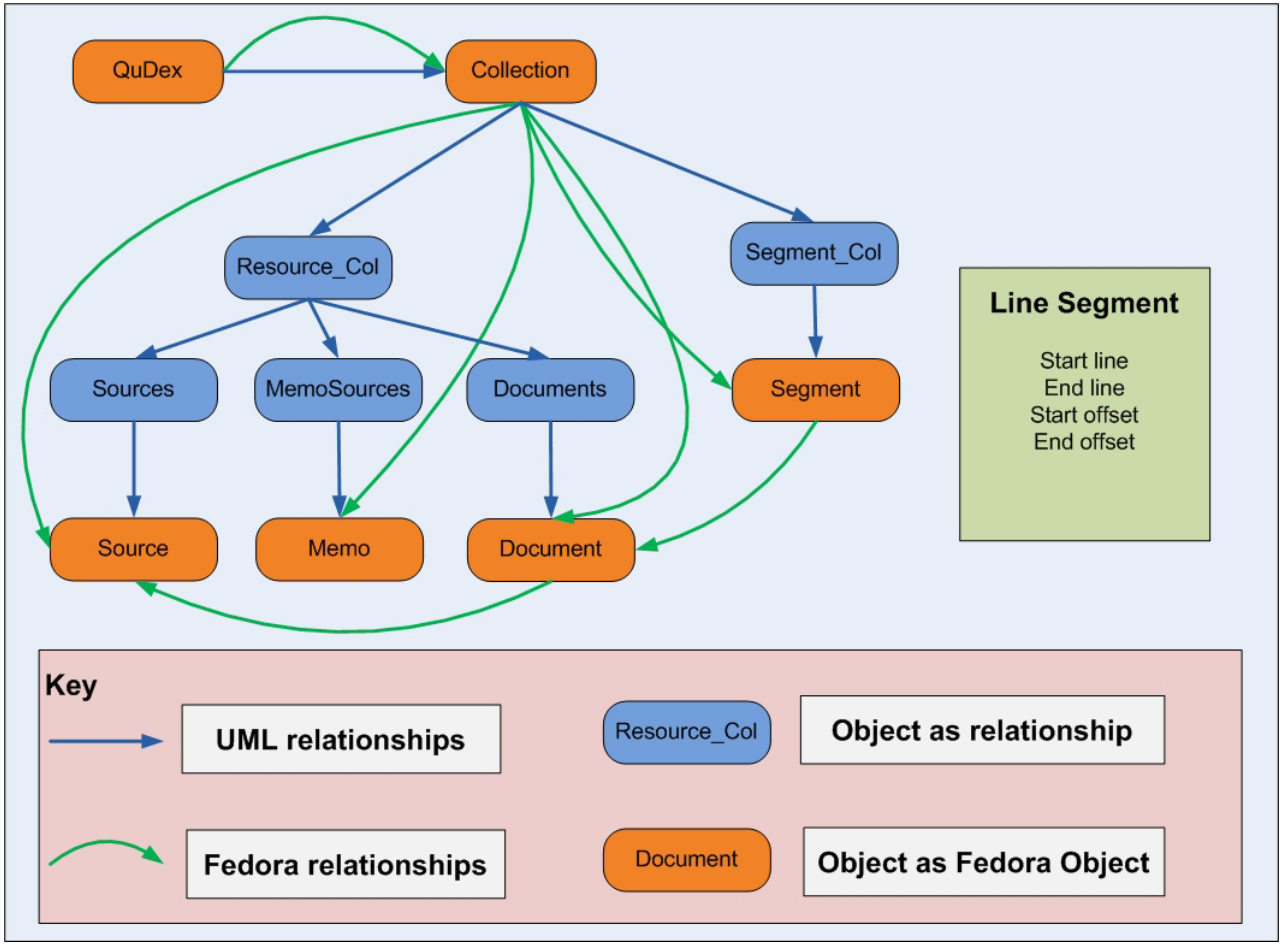
Modelling QuDeX



Objects from the UML model are translated into either Fedora Objects, Relationships or Data streams, or a combination depending on the type of UML object and its attributes and/or stereotypes

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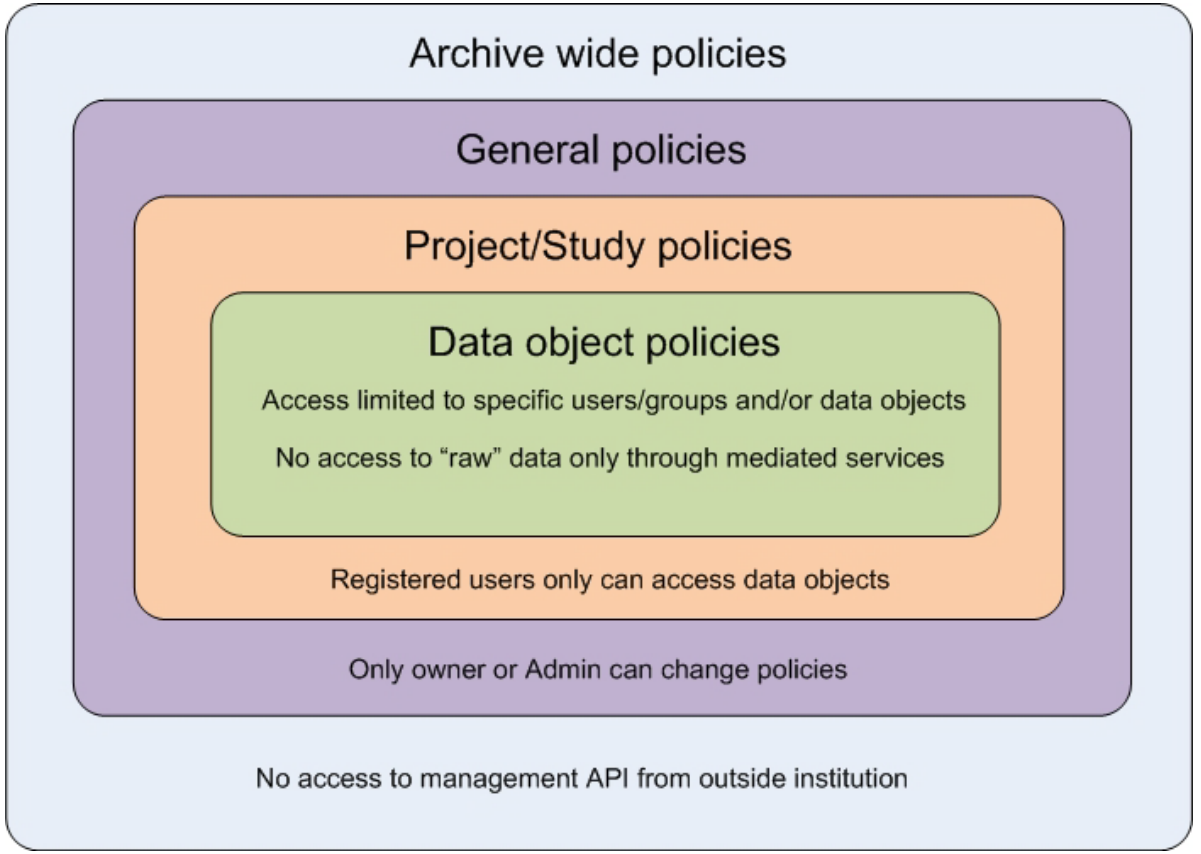
Authentication and Authorization



Policy based authentication and authorization using XACML

Policies go from the general to the specific

Policies can be object based where they can be **versioned**

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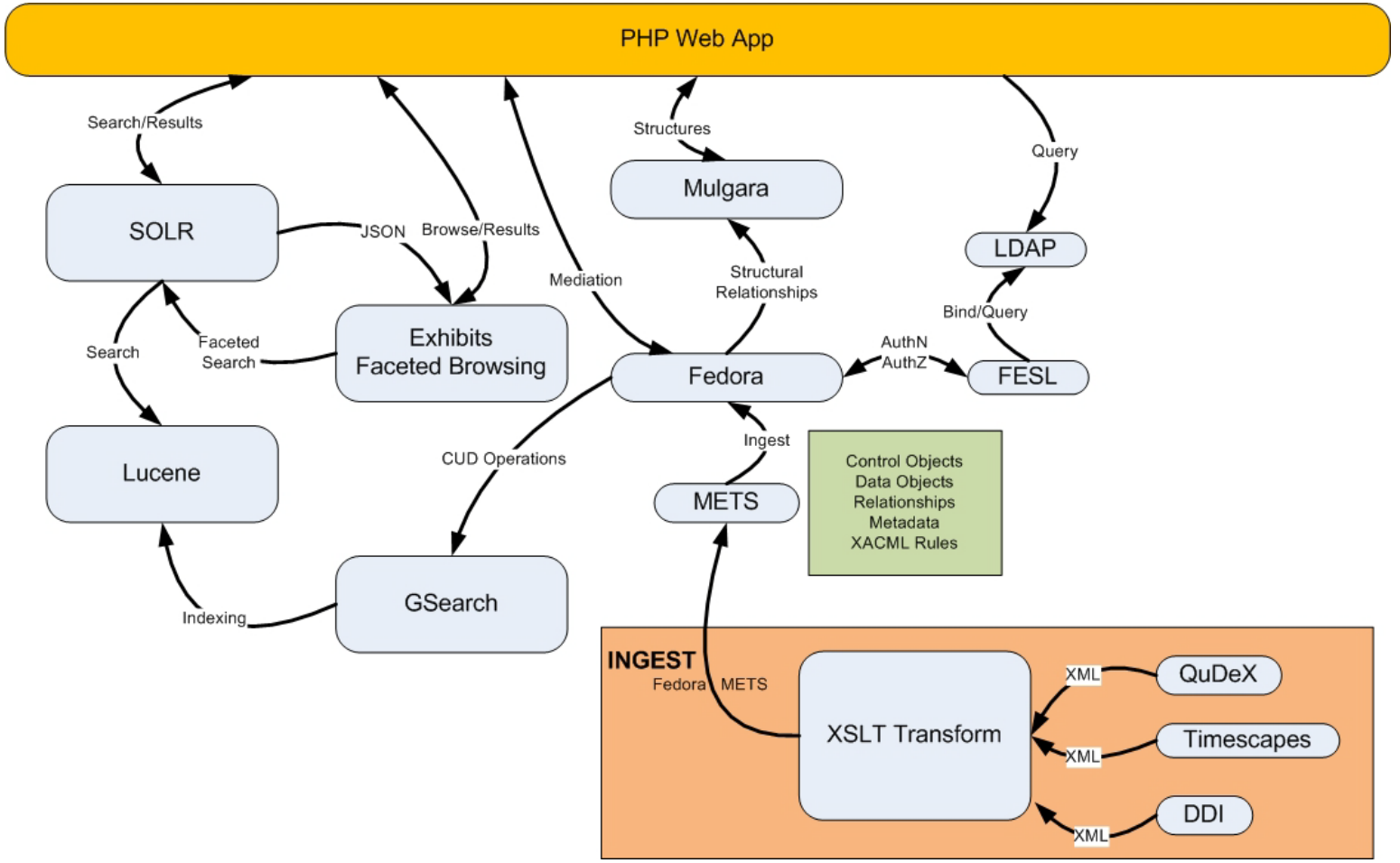
The system